

Day 10 Hybrid Notes

Altitude relationships and geometric mean as support | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can recognize similar triangles formed by an altitude to the hypotenuse.
I can use proportions to solve right-triangle similarity problems.
I can connect geometric mean relationships to proportional reasoning.

Part 1: Cloze Notes - Essential Ideas
1. An altitude to the hypotenuse of a right triangle creates smaller triangles that are _____.
2. Similar right triangles have congruent corresponding _____.
3. Corresponding side lengths in similar triangles are _____.
4. A geometric mean is found between two positive _____.
5. In a proportion $a/b = c/d$, cross products are _____.
6. The hypotenuse is opposite the right _____.
7. Always match sides before writing a _____.

Part 2: Outline Notes - Big Picture A. Right-triangle similarity is a bridge between similarity and _____. B. The altitude creates three related _____. C. Geometric mean formulas are shortcuts for _____. D. Weekly review combines quadrilaterals, similarity, and _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 10 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. Solve the proportion $6/x = x/24$.

Setup/work: _____

Answer: _____

2. If a small right triangle is similar to a large one with scale factor 4, a side of 5 becomes what?

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student writes $6/x = 24/x$ for a geometric mean problem. What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A ladder diagram creates similar right triangles. A 3-ft side corresponds to a 12-ft side. If another small side is 5 ft, find the matching large side.

Setup: _____

Answer: _____

Sentence answer: _____